Faculty of Computing & Information Technology

CC-112L : Programming Fundamentals **Lab Instructor**: Dr. Muhammad Farooq

BSAI/IT Morning/Self 2024
Quiz – 02 solution Total: 3+5+12+5+10= 35 **Date : 16/Monday /2026**

The objective of this quiz is to:

- Pseudocode
- Nested if-else
- Operators, Arithmetic

MCQS

MCQS

Question 1: Path Selection

Answer: B) High Heat

Reason:
Temperature = 35
Humidity = 75

35 > 30 so first block runs.
75 is not greater than 80 so High Heat is displayed.

Question 2: Deep Tracing

Answer: B) Green

Reason:
X = 10
Y = 20

10 < 15 so outer condition runs.
20 is not greater than 25 so next condition is checked.
10 + 20 = 30 so Green is displayed.

Question 3: Hierarchy of Execution

Answer: C)

If outer condition fails the inner block is skipped and control moves forward.

TRUE FALSE SECTION

Question 1: The ATM Withdrawal

Answer: FALSE

Reason:

Input equals PIN correct.
1000 <= 1200 correct.
1000 <= 500 not correct.

Question 2: E Commerce Shipping

Answer: TRUE

Reason:
Location is International.
200 > 150 so condition satisfied.

Question 3: Student Grading

Answer: TRUE

Reason:
88 + 5 = 93.
93 >= 90 satisfied.
98 > 95 satisfied.

Question 4: Smart Lighting

Answer: TRUE

Reason:
Hour is 5.
5 <= 6 so night condition satisfied.

Question 5: Library Access

Answer: FALSE

Reason:
Member is Regular not Gold.
First part fails.

PSEUDOCODE SECTION

Question 1: The Automated ATM Withdrawal

Answer:

```
BEGIN
DECLARE StoredPIN AS INTEGER
```

```
DECLARE EnteredPIN AS INTEGER
DECLARE WithdrawalAmount AS INTEGER
DECLARE AccountBalance AS INTEGER
INPUT StoredPIN
INPUT EnteredPIN
INPUT WithdrawalAmount
INPUT AccountBalance
IF EnteredPIN == StoredPIN THEN
   IF WithdrawalAmount <= AccountBalance THEN
      DISPLAY "Withdrawal Successful"
   ELSE
      DISPLAY "Insufficient Funds"
ELSE
   DISPLAY "Invalid PIN"
END
```

Question 2: E Commerce Shipping Calculator

Answer:

```
BEGIN
DECLARE Location AS STRING
DECLARE OrderTotal AS INTEGER
INPUT Location
INPUT OrderTotal
IF Location == "Local" THEN
   IF OrderTotal > 50 THEN
      DISPLAY "Free"
   ELSE
      DISPLAY "$5"
ELSE IF Location == "International" THEN
   IF OrderTotal > 150 THEN
      DISPLAY "$15"
   ELSE
      DISPLAY "$30"
END
```

Question 3: Smart Home Lighting System

Answer:

```
BEGIN
DECLARE Motion AS STRING
DECLARE Hour AS INTEGER
DECLARE Brightness AS STRING
INPUT Motion
INPUT Hour
IF Motion == "Detected" THEN
   IF Hour >= 22 OR Hour <= 6 THEN
      SET Brightness = "30%"
   ELSE
      SET Brightness = "100%"
ELSE
   SET Brightness = "OFF"
DISPLAY Brightness
END
```

Question 4: Library Book Loan System

Answer:

```
BEGIN
DECLARE MemberType AS STRING
DECLARE Fines AS INTEGER
INPUT MemberType
INPUT Fines
IF MemberType == "Gold" THEN
   IF Fines > 0 THEN
      DISPLAY "Clear fines to borrow"
   ELSE
      DISPLAY "Borrowing allowed"
ELSE IF MemberType == "Regular" THEN
   DISPLAY "Upgrade to Gold to borrow"
ELSE
   DISPLAY "Please sign up for a membership"
END
```

Question: University Academic Performance Criteria

Answer:

```
BEGIN
DECLARE Score AS INTEGER
DECLARE Attendance AS INTEGER
INPUT Score
INPUT Attendance
IF Score >= 90 THEN
   IF Attendance > 95 THEN
      DISPLAY "Grade A with Honors"
   ELSE
      DISPLAY "Grade A"
ELSE IF Score >= 50 THEN
   DISPLAY "Pass"
ELSE
   DISPLAY "Fail"
END
```

Question: Multi Zone Smart Climate Control System

Answer:

```
BEGIN
DECLARE Occupancy AS STRING
DECLARE WeatherAlert AS STRING
DECLARE Zone AS STRING
DECLARE Time AS INTEGER
DECLARE EnergySaver AS STRING
DECLARE Winter AS STRING
DECLARE Temperature AS INTEGER
INPUT Occupancy
INPUT WeatherAlert
INPUT Zone
INPUT Time
INPUT EnergySaver
INPUT Winter
```

```
IF Occupancy == "Triggered" THEN
   IF WeatherAlert == "Active" THEN
      IF Zone == "Lab" THEN
         SET Temperature = 22
      ELSE IF Zone == "Office" THEN
         SET Temperature = 24
   ELSE
      IF Time >= 8 AND Time <= 18 THEN
         IF EnergySaver == "ON" THEN
            SET Temperature = 25
         ELSE
            SET Temperature = 21
      ELSE
         SET Temperature = 26
ELSE
   IF Winter == "Yes" THEN
      SET Temperature = 15
   ELSE
      SET Temperature = 0
DISPLAY Temperature
END
```